IE 643 COURSE PROJECT · WORKING ROADMAP

Probing the Latent Space of Video Diffusion U-Nets

Disentangling Content vs. Motion

Team Claudn't · revised 21 September 2026 · supersedes the earlier draft

1. Project snapshot

Team: Claudn't · **Roll numbers:** [fill in, one per member]

The task. A video diffusion U-Net does two jobs in alternation. Its spatial layers work inside a single frame and decide what the thing looks like; its temporal layers work across frames and decide how it moves. Our job is to find out where that division actually sits inside the network, which layers and which denoising timesteps, and then prove we found it by making an edit that only that understanding would allow. The target demonstration is a clip of a golden retriever running on a beach where we keep the exact running gait but turn the dog into a cat, and the reverse, where the dog stays but starts walking instead.

Why this is a probing project and not a generation project. We are not trying to build a better video model. We are trying to explain one. The edit at the end is evidence, not the product. That framing matters for how we spend compute: a careful sweep over a small, controlled clip set beats a large, messy one.

Timeline. Course staff extended the overall project by roughly a month and a half, which puts the intensive assessment somewhere in early-to-mid November 2026. We have not seen a firm date yet, so everything past 1 November in the table below is provisional. The prep presentation deadline (21 Sep) may or may not have moved with it; we are treating it as still live and submitting on time.

2. Where we are: 2 to 20 September

We started on 2 September. The window from 11 to 20 September ran into midsems, so we deliberately planned almost nothing for it apart from reading, and that is how it went. Everything below is what actually happened in those three weeks.

2.1 Reading completed

Groups 1 to 5 of the reading list in section 3 are done. Group 7 (the model and tooling documentation) we are reading as we build, which is the sensible order for it. The single most useful thing we read was not on the original list at all: a small cluster of recent papers that have already gone looking for motion inside video diffusion features. They changed what we plan to measure. See section 5.2.

2.2 Pilot experiments

Four short experiments, run on a single GPU, mostly to find out whether the project is tractable at all before committing to a method.

ID	Question we were asking	What we ran	Outcome
E1	Can we generate video at all on the hardware we have?	AnimateDiff v1.5 (motion adapter v1-5-2) on an SD1.5 backbone through `diffusers`. 16 frames, 512×512, 25 DDIM steps.	Works. [Fill in: wall-clock per clip ___ s, peak VRAM ___ GB.] This sets the budget for every sweep we plan later.

E2	If we fix the seed and change one word of the prompt, does the rest of the clip stay put?	Same latent noise, same settings, prompt varied along one axis at a time: noun swap (retriever → cat) and verb swap (running → walking).	Largely yes. Composition and camera framing survive a fixed seed well enough that paired clips are a usable controlled design. Not perfect, and the failure cases are informative: a noun swap between very differently shaped animals drags the layout with it.
E3	Can we get activations out, and how much disk does that cost?	Forward hooks on `unet.up_blocks[i].motion_modules[j]` and on the `attn2` cross-attention modules. Dumped shapes and sizes for one clip.	Straightforward to hook. Storage is the real constraint, not compute. [Fill in: ___ GB per clip at fp16 for a full dump.] Conclusion: we cannot store everything, so the harness has to store summaries by default and full tensors only on request.
E4	Do the temporal modules causally carry motion, or is that just an architectural story?	Scaled the motion-module residual contribution down towards zero and regenerated with an otherwise identical configuration.	The clip collapses towards a near-static sequence while each individual frame stays a plausible image. That is the cheapest possible causal demonstration that these modules are the cross-frame path, and it is a good opening result for the report.

2.3 What we decided because of the pilots

- AnimateDiff is the primary model. E1 settled the feasibility question and E4 confirmed the motion module is a real causal handle, not just a label on a box.
- Every clip in the probing set will be generated from a fixed seed and a fixed noise tensor. E2 is the justification. Without it, activation differences between two clips are dominated by sampling noise and the whole probe is meaningless.
- The harness stores per-coordinate summary statistics by default. E3 killed the original plan of dumping every activation to disk.

3. Reading roadmap

Grouped by subtopic, roughly in the order we read them. Citations follow the course format. Access dates for the two websites need filling in with the actual date you consulted them.

Group 1. Video diffusion foundations

- Ho, Jonathan, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi and David J. Fleet. Video Diffusion Models, NeurIPS, 2022. https://arxiv.org/abs/2204.03458
- Blattmann, Andreas, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler and Karsten Kreis. Align your Latents: High-Resolution Video Synthesis with Latent Diffusion Models, CVPR, 2023. https://arxiv.org/abs/2304.08818
- Blattmann, Andreas, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, Varun Jampani and Robin Rombach. Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets, arXiv, 2023. https://arxiv.org/abs/2311.15127

These establish the factorisation we are probing: spatial blocks inside a frame, temporal blocks across frames, repeated at every resolution.

Group 2. Motion modules and separation by design

- Guo, Yuwei, Ceyuan Yang, Anyi Rao, Zhengyang Liang, Yaohui Wang, Yu Qiao, Maneesh Agrawala, Dahua Lin and Bo Dai. AnimateDiff: Animate Your Personalized Text-to-Image Diffusion Models without Specific Tuning, ICLR, 2024. https://arxiv.org/abs/2307.04725
- Zhao, Rui, Yuchao Gu, Jay Zhangjie Wu, David Junhao Zhang, Jiawei Liu, Weijia Wu, Jussi Keppo and Mike Zheng Shou. MotionDirector: Motion Customization of Text-to-Video Diffusion Models, ECCV, 2024. https://arxiv.org/abs/2310.08465
- Wu, Jay Zhangjie, Yixiao Ge, Xintao Wang, Stan Weixian Lei, Yuchao Gu, Yufei Shi, Wynne Hsu, Ying Shan, Xiaohu Qie and Mike Zheng Shou. Tune-A-Video: One-Shot Tuning of Image Diffusion Models for Text-to-Video Generation, ICCV, 2023. https://arxiv.org/abs/2212.11565

Closest prior art to our deliverable. MotionDirector's dual-path appearance/motion LoRA is the design we compare against; it shows the split is exploitable, which is a weaker claim than ours but the same territory.

Group 3. Activation patching as a method

- Meng, Kevin, David Bau, Alex Andonian and Yonatan Belinkov. Locating and Editing Factual Associations in GPT, NeurIPS, 2022. https://arxiv.org/abs/2202.05262
- Wang, Kevin, Alexandre Variengien, Arthur Conmy, Buck Shlegeris and Jacob Steinhardt. Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 Small, ICLR, 2023. https://arxiv.org/abs/2211.00593
- Zhang, Fred and Neel Nanda. Towards Best Practices of Activation Patching in Language Models: Metrics and Methods, ICLR, 2024. https://arxiv.org/abs/2309.16042

Developed on language models, but the causal logic carries over unchanged. Read Zhang and Nanda properly before writing any patching code; their warnings about metric choice and about noising versus resampling corruption are exactly the mistakes we would otherwise make.

Group 4. Cross-attention as a diagnostic

- Hertz, Amir, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch and Daniel Cohen-Or. Prompt-to-Prompt Image Editing with Cross Attention Control, ICLR, 2023. https://arxiv.org/abs/2208.01626
- Tang, Raphael, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Jimmy Lin and Ferhan Ture. What the DAAM: Interpreting Stable Diffusion Using Cross Attention, ACL, 2023. https://arxiv.org/abs/2210.04885

Working recipes for pulling attention maps out of a diffusion U-Net per layer and per timestep. We adapt DAAM's aggregation from one image to a stack of frames.

Group 5. Where motion has already been located (added after the pilots)

- Xiao, Zeqi, Yifan Zhou, Shuai Yang and Xingang Pan. Video Diffusion Models are Training-free Motion Interpreter and Controller, NeurIPS, 2024. https://arxiv.org/abs/2405.14864
- Yatim, Danah, Rafail Fridman, Omer Bar-Tal, Yoni Kasten and Tali Dekel. Space-Time Diffusion Features for Zero-Shot Text-Driven Motion Transfer, CVPR, 2024. https://arxiv.org/abs/2311.17009
- Ling, Pengyang, Jiazi Bu, Pan Zhang, Xiaoyi Dong, Yuhang Zang, Tong Wu, Huaian Chen, Jiaqi Wang and Yi Jin. MotionClone: Training-Free Motion Cloning for Controllable Video Generation, ICLR, 2025. https://arxiv.org/abs/2406.05338
- Baherwani, Vatsal, Yixuan Ren and Abhinav Shrivastava. Characterizing Motion Encoding in Video Diffusion Timesteps, arXiv, 2025. https://arxiv.org/abs/2512.22175
- Kwon, Mingi, Jaeseok Jeong and Youngjung Uh. Diffusion Models already have a Semantic Latent Space, ICLR, 2023. https://arxiv.org/abs/2210.10960

This group is the reason our plan changed. Together these papers tell us roughly where to look before we start looking, which turns a blind sweep into a set of testable predictions. Details in section 5.2.

Group 6. Earlier framings of the same split

- Tulyakov, Sergey, Ming-Yu Liu, Xiaodong Yang and Jan Kautz. MoCoGAN: Decomposing Motion and Content for Video Generation, CVPR, 2018. https://arxiv.org/abs/1707.04993
- Yu, Sihyun, Weili Nie, De-An Huang, Boyi Li, Jinwoo Shin and Anima Anandkumar. Efficient Video Diffusion Models via Content-Frame Motion-Latent Decomposition, ICLR, 2024. https://arxiv.org/abs/2403.14148

MoCoGAN is where the content/motion vocabulary in our project brief comes from, and CMD is the modern diffusion-era version of the same idea. Both build the split in by construction. We are asking the harder question: does a model that was never factored this way learn a version of it anyway?

Group 7. Tools and documentation (read while building)

- Hugging Face. Diffusers documentation, text-to-video pipelines. https://huggingface.co/docs/diffusers/using-diffusers/text-img2vid. Accessed on: [fill in].
- Guo, Yuwei et al. AnimateDiff model card. https://huggingface.co/guoyww/animatediff. Accessed on: [fill in].
- Teed, Zachary and Jia Deng. RAFT: Recurrent All-Pairs Field Transforms for Optical Flow, ECCV, 2020. https://arxiv.org/abs/2003.12039
- Karaev, Nikita, Ignacio Rocco, Benjamin Graham, Natalia Neverova, Andrea Vedaldi and Christian Rupprecht. CoTracker: It is Better to Track Together, ECCV, 2024. https://arxiv.org/abs/2307.07635
- Huang, Ziqi, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao and Ziwei Liu. VBench: Comprehensive Benchmark Suite for Video Generative Models, CVPR, 2024. https://arxiv.org/abs/2311.17982

4. Model and dataset shortlist

Primary model: AnimateDiff. An SD1.5 UNet (roughly 860M parameters, frozen) with a separately trained motion module inserted after its residual and attention blocks. The motion module checkpoint `mm_sd_v15_v2` is about 417M parameters, around 1.6 GB on disk. The earlier draft of this document said the motion module was 1.7B; that was wrong, it confused the file size in gigabytes with a parameter count. The total is closer to 1.3B parameters before the VAE and text encoder.

We picked it because the spatial/temporal division is already explicit in the code, which means Deliverable 2 starts from a hypothesis rather than a blank sheet. The honest caveat, and it belongs in the report: because the motion module was trained on top of a frozen image model, the "content" layers never saw motion during training. Any separation we find is partly an artefact of that training procedure. This is exactly why the ModelScope check matters.

Model	Size	Why it is on the list	Why it is not primary
AnimateDiff primary	417M motion module + ~860M SD1.5 UNet	Explicit spatial/temporal split in the module tree; runs on one 12–16 GB GPU; mature `diffusers` support	Motion module bolted onto a frozen image model, so the split is partly given rather than learned jointly
ModelScope T2V `damo-vilab/text-to-video-ms-1.7b`	1.7B	Spatial and temporal trained together end to end. The harder and more interesting test case	Thinner tooling for attention extraction than SD-derived pipelines

Stable Video Diffusion (img2vid)	~1.5B UNet	Best output quality of the three, well documented	Image-conditioned only, so we cannot build matched content/ motion pairs from text prompts

Plan: all of Deliverables 1 to 4 on AnimateDiff. If Stage 7 has room, repeat the Deliverable 2 sweep only on ModelScope T2V as a robustness check. Deliverable 4 asks us to demonstrate disentanglement, not to demonstrate it twice, so this is a bonus rather than a requirement.

Probing data

We generate our own clips. That is not a shortcut; it is the only way to get pairs that differ along exactly one axis, and without that the probe has nothing to measure.

- **The core paired set.** A prompt grid of the form *[subject] [verb] [scene]*. Hold two of the three fixed, vary the third. A content pair changes only the subject; a motion pair changes only the verb. Same seed, same noise tensor, same sampler settings across a pair. Target size is around 120 clips, roughly 15 subjects by 4 verbs by 2 scenes, generated once and reused everywhere.
- **Prompt realism.** Subject and scene vocabulary sampled from MSR-VTT and WebVid-10M captions so we are not testing on a handful of toy prompts. WebVid in particular is close to what these models were pretrained on.
- **Motion vocabulary.** Verbs taken from UCF101 and Kinetics-400 action labels. They give us a clean, standard list of distinct motions (running, walking, jumping, spinning) instead of whatever we happened to think of.

We are not training on any of these datasets. They are a source of vocabulary and structure only. Generation time is the bottleneck here, not data availability, which is why the set stays small and controlled.

5. Refined plan, deliverable by deliverable

This section is where the reading in group 5 changed things. The original plan was to sweep every layer and every timestep and look for structure. That is still the shape of it, but we now know enough to predict what we should find, which makes the result falsifiable instead of merely descriptive.

5.1 Deliverable 1: setup and the extraction harness

- **Hook targets.** In `diffusers`, AnimateDiff builds a `UNetMotionModel` from a `UNet2DConditionModel` plus a `MotionAdapter`. The temporal path is the `motion_modules` list inside each `down_blocks[i]`, `up_blocks[i]` and the mid block. The spatial path is the `transformer_blocks`, with `attn1` self-attention and `attn2` cross-attention to the text embedding. Forward hooks on both, keyed by module path.
- **Coordinate system.** Everything is indexed by a triple: (module path, denoising timestep, clip id). Every later stage queries by that triple, so getting it right once saves a lot of pain.
- **What gets stored.** Not the raw tensors, by default. E3 showed a full dump is impractical. Instead we store a per-coordinate summary (channel means and variances, frame-mean-removed norms, attention maps downsampled to a fixed grid) and keep full tensors only for a short list of coordinates we mark as interesting.
- **Determinism.** One seed policy, written down and enforced in the generation script rather than passed by hand. Every clip records the seed, sampler, step count, resolution and model revision alongside its activations.
- **Attention maps.** DAAM-style aggregation, extended to video by keeping the frame axis rather than collapsing it. A token's map becomes a small movie, which is what makes it useful for telling content attention apart from motion attention.

5.2 Deliverable 2: three predictions, then a measurement

Rather than sweeping blind, we go in with three hypotheses drawn from the group 5 papers. Each is falsifiable and each is cheap to test with the harness from Deliverable 1.

H1, the layer axis. Motion information concentrates in the temporal modules at medium resolution rather than at the extremes. Xiao et al. find their motion feature works best after the first up-block and report that mid-resolution features give better motion correspondence than either shallow or deep ones. Prediction: our motion selectivity peaks around `up_blocks[1]`, not at the highest-resolution decoder layers and not at the bottleneck.

H2, the timestep axis. Motion is decided early, appearance late. This is a widely repeated piece of folklore that Baherwani et al. actually went and tested, finding two regimes with a reasonably consistent boundary between them. Prediction: motion selectivity is high in the first third of the denoising trajectory and falls off; content selectivity does the opposite. If the boundary sits somewhere different for AnimateDiff than they report, that is a result worth writing up, not a bug.

H3, the content confound. Raw activations, even in temporal layers, are dominated by appearance. Xiao et al. only get a clean motion signal after subtracting the per-frame mean across the frame axis, which removes the static component. Prediction: if we skip that normalisation, every layer will look content-selective and we will wrongly conclude there is no motion anywhere. This one matters most, because it is the mistake we would otherwise have made in week one of the sweep.

The measurement. For each coordinate *(layer L, timestep t)*, take a content pair and a motion pair that share a seed. Let d_c be the distance between cached activations under the content swap and d_m the distance under the motion swap. Define

$$S(L, t) = (d_c - d_m) / (d_c + d_m)$$

which lands in the range −1 to 1. Positive means the coordinate responds to what is depicted, negative means it responds to how things move, near zero means it responds to both or to neither. Averaged over pairs, this gives one heat map with layers down the rows and timesteps across the columns, and it is the central figure of the whole project. It is a better object than raw divergence because the sign carries the meaning.

The correlational counterpart. Separately, fit a small linear probe on each coordinate's cached activation to predict the motion verb and, in a second probe, the subject noun. Probe accuracy per coordinate says what information is *present*. The patching in Deliverable 3 says what information is *used*. These two can disagree, and where they disagree is the most interesting thing we are likely to find.

5.3 Deliverable 3: the patching framework

A patch replaces the activation at a chosen coordinate during clip B's generation with the one cached from clip A. Both runs must share an initial noise tensor or the comparison is meaningless.

Four granularities, in increasing order of precision. We implement them in this order and stop when one works well enough:

1. **Whole tensor.** Replace the full activation. Blunt, and likely to break coherence, but it establishes an upper bound on effect size.
2. **Channel subset.** Replace only the channels with the highest motion selectivity, following the top-few-percent channel filtering in Xiao et al. We expect this is where the method starts working.
3. **Attention weights only.** Replace the temporal attention map rather than the features, in the spirit of MotionClone's observation that the dominant components of temporal attention are what actually drive motion.
4. **Scaled residual injection.** Blend rather than replace, with a strength α from 0 to 1. Sweeping α gives a dose-response curve, which is far stronger evidence than a binary worked/did-not-work.

Three controls, all of which we run before believing any positive result:

- **Null patch.** Patch a clip with its own activation. The output must come back identical, or near enough. If it does not, there is a bug in the harness and nothing downstream is trustworthy.
- **Random coordinate.** Patch a coordinate with low selectivity. The effect should be small. If a random layer does as well as our chosen one, we have not localised anything.
- **Symmetry.** Patch A into B and B into A. The two effects should be approximate inverses. Asymmetry means something is leaking.

5.4 Deliverable 4: the demonstration, and how we score it

Two edits per test case, each with a clear target: keep A's motion while taking B's content, and the reverse. Reported as a video grid plus numbers.

Content fidelity. CLIP similarity between output frames and the intended content prompt, plus CLIP or DINO image-to-image similarity against the reference clip's frames. The image-to-image version is the more honest of the two, since CLIP text similarity is easy to satisfy accidentally.

Motion fidelity. A trajectory-based score in the style of Yatim et al.: extract point tracks from both clips with an off-the-shelf tracker (CoTracker) and compare the displacement vectors over time. Optical flow correlation via RAFT as a cheaper second opinion.

Video quality. Two or three relevant VBench dimensions (subject consistency, motion smoothness, temporal flickering) so we can show the edit did not simply wreck the clip. Cheap to run, and it pre-empts the obvious objection.

The headline number. A single disentanglement score built from the four cross-similarities of one edit. For an output that should carry A's motion and B's content:

$$D = \tfrac{1}{2}[\, s_c(\text{out},B) + s_m(\text{out},A) \,] - \tfrac{1}{2}[\, s_c(\text{out},A) + s_m(\text{out},B) \,]$$

s_c is content similarity, s_m is motion similarity, out is the edited clip.

Positive and large means the edit moved the axis we aimed at and left the other one alone. Reported separately for both edit directions, because a method that works one way and not the other has not demonstrated disentanglement.

Baselines. This was the biggest gap in the previous draft. A disentanglement score with nothing to compare against is an unanchored number. At minimum we run:

- **Prompt-only.** Just generate with the new prompt and the same seed, no patching at all. If our intervention cannot beat this, we have built an expensive way to do nothing. This one is mandatory.
- **Training-free reference.** MotionClone or the MOFT controller, both of which have public code. Comparable in spirit to what we are doing.
- **Trained reference.** MotionDirector, if time allows. This is the ceiling we do not expect to reach, and saying so plainly is better than pretending otherwise.

5.5 The "trained/finetuned model" requirement

Our method is mostly training-free, which sits awkwardly with a deliverable that asks for a trained model. The clean way to satisfy it, and to get something useful out of it, is a small MotionDirector-style pair of LoRAs: one appearance LoRA and one motion LoRA, fit on a handful of clips. Two reasons to do it. It gives us a reference disentanglement produced by training rather than by patching, so we can ask whether our patched coordinates line up with what the LoRAs learned. And it gives the submission an actual set of weights to hand in alongside the base checkpoint. Budget one or two days in Stage 6, not more.

5.6 Interface

A Gradio app. The important design decision is that it should expose the interpretability rather than hide it.

Inputs: two prompts (or two generated clips), a layer selector, a timestep range, and the patch strength α. Outputs: the edited clip, the three metric values, and the cross-attention heat map for the selected coordinate shown beside the selectivity map from Deliverable 2, so a user can see *why* that coordinate was chosen. A demo that only shows the output video would satisfy the requirement, but it would waste the project. Confirm the expected scope with the TA regardless; the brief explicitly says to ask.

6. Week-by-week roadmap

Stage 0 and the midsem window are done. Everything from Stage 1 onwards is plan. Dates past 1 November depend on the assessment date being confirmed.

Stage	Dates	Focus	Maps to
0	Sep 2 – Sep 10	Scoping and framing. Reading groups 1–3. Environment and model setup. Pilot experiments E1–E4. Experiment log started.	Deliverable 1 groundwork
—	Sep 11 – Sep 20	**Midsem window, deliberately light.** Reading groups 4–5 only, a few hours across the ten days. No compute work and no deadlines set in this period.	Background only
1	Sep 21 – Sep 27	Record and submit the prep presentation. Read group 7 alongside. Freeze the generation protocol: seed policy, resolution, frame count, step count, scheduler.	Prep submission
2	Sep 28 – Oct 4	Build the extraction harness properly. Hooks on all spatial and temporal blocks, per-timestep attention dumps, the (module, timestep, clip) index, the summary-statistics store. Generate the first 40 clips of the paired set.	Deliverable 1
3	Oct 5 – Oct 11	Finish the paired set. Run the selectivity sweep on a coarse grid. Test H1 and H3 first, since H3 determines whether any of the numbers mean anything.	Deliverable 2
4	Oct 12 – Oct 18	Refine the grid around whatever Stage 3 flagged. Test H2 across the timestep axis. Fit the linear probes. Produce the layer-by-timestep selectivity map and write up Stage 2 of the report while it is fresh.	Deliverable 2
5	Oct 19 – Oct 25	Patching framework. Granularities 1 and 2 first, then 3 and 4 if needed. All three controls. Sweep α for the dose-response curve.	Deliverable 3
6	Oct 26 – Nov 1	Edit demos in both directions across the test cases. Compute all metrics against the prompt-only baseline at minimum. Train the appearance/motion LoRA pair. Collect the qualitative grids.	Deliverable 4
7	Nov 2 – assessment	Gradio interface. Stage-wise report. Package weights, code, README, environment file. Optional ModelScope robustness check if the schedule holds. Rehearse.	Final package

Where the slack is. Video inference and hook-based extraction are slower and more fragile than the equivalent image-diffusion work, and Stage 2 is the one most likely to overrun. If it does, take the time out of Stage 5, not Stage 3. The Stage 3 and 4 probing results are what everything else is built on; a rushed patching implementation is recoverable, a rushed sweep is not.

Standing habit. The experiment log started in Stage 0 stays running to the end. Date, what was tried, model and layer configuration, result, next step. It is dull to maintain and it is the only reason the stage-wise report will be writable in Stage 7 instead of reconstructed from memory.

7. Final deliverables checklist

Six items are due at the intensive assessment, separately from the prep presentation.

- **Trained or finetuned model.** The appearance/motion LoRA pair from Stage 6, plus the exact base checkpoint and revision hashes for AnimateDiff and the SD1.5 backbone.
- **Code with documentation.** Extraction harness, patching framework, metrics, interface. A README covering setup, dependencies (`torch`, `diffusers`, checkpoint sources), and one command per figure so each result can be reproduced.
- **Interface.** The Gradio app described in 5.6. Confirm the expected scope with the TA early, since this varies a lot by project.
- **Performance metrics.** Content fidelity, motion fidelity, the disentanglement score D, and at least the prompt-only baseline to compare against. Both edit directions reported.
- **Qualitative results.** Side-by-side grids: original A, original B, content of A with motion of B, and the reverse. Non-negotiable given the project is visual.
- **Stage-wise report.** Why each step was taken, the major experiments and what they showed (including the ones that failed), how the probing set was built, and the interface methodology. More detailed instructions are promised by the course; the experiment log is what makes this writable.

8. Prep presentation and video checklist

Slides, against the course instruction sheet:

- 25 slides maximum. The companion deck is at 25 exactly, so any addition needs a deletion.
- Title slide with the topic, an explicit statement that this is IE 643 course project work, the team name and every member's roll number. **Roll numbers still need filling in.**
- Outline slide.
- Background materials discussed rather than listed.
- Approaches tried so far and the future plan, broken into subtasks.
- Diagrams for the hard concepts: the U-Net factorisation, the patching mechanism, the selectivity map.
- Data collection and curation described concretely.
- Models investigated and how each will be used.
- Pretraining data of the models used, plus our own probing set and why it is built the way it is.
- Experiments carried out so far. E1 to E4 are on the deck; **fill in the measured numbers from your own runs.**
- References in the exact required format.

Video:

- 15 minutes maximum. There is a penalty for going over, so rehearse against a timer rather than a feeling.
- Every team member presents and is on camera throughout, not only during their own section.
- Do not read the slides out. The instruction sheet says this twice and it is the easiest mark to lose.
- OBS Studio for recording (https://obsproject.com/wiki/install-instructions). Scene with slide capture plus a webcam overlay. Do one full test take before the real one and check audio levels and lighting on it.

9. Submission

- Google Drive folder named exactly `Claudnt_IE643_CourseProject_Prep`, created with IITB email and SSO. The team name has an apostrophe in it, which Drive and the course form may not handle cleanly, so drop it in the folder name and keep the real spelling on the slides.
- Both the slides and the video go in that folder.
- Share it so the instructor and every TA can open it. Test in an incognito window rather than assuming. Inaccessible content is not considered submitted.
- Submit the folder link through the course form once it is posted.
- Confirm the revised prep presentation deadline separately from the assessment date. The two were not necessarily moved together.

10. Open questions for the TA

- Firm dates for the prep presentation and for the intensive assessment. The note only said "about a month and a half".
- What counts as an acceptable interface here. A live Gradio app, or would an annotated notebook with input cells do? The brief tells us to ask.
- Is one primary model enough for full credit, or is cross-model validation expected? This changes how much of Stage 7 we reserve.
- Is GPU time provided through the course, or do we source our own? This sets the resolution of the Stage 3 sweep more than any other single factor.
- Given that our method is largely training-free, does the LoRA pair described in 5.5 satisfy the "trained/finetuned model" deliverable?
- Any required structure or rubric for the stage-wise report beyond what has been announced.